

재료의 기계적 거동(공) (MSE407-01)

학기	2026년 1학기
담당교수	김영규
이수구분	교선평수
학점	3.0
강의시간	월13:30-15:00(E7 - L20), 수13:30-15:00(E7 - L20)
강의실	E6 210 (분석실 Ⅲ)
소속	기초학부

교과개요

This course provides the theoretical foundation needed to understand and control the mechanical behaviors of materials as a material scientist and/or engineer.

교과목표

After attending this course, you will be able to correlate materials' atomistic behaviors to their macroscopic responses. We begin by establishing the fundamental mathematical background to describe mechanical responses and explaining elastic behaviors from an atomic point of view. We then transition to permanent deformation, starting with the concepts of plasticity and yield criteria. We will understand plastic deformation process through the lens of crystallography, as we discuss slips and impact of dislocations. Next, we study critical failure mechanisms such as creep, fracture, and fatigue by looking into both theory and real life examples. The course concludes with practical case studies on metals, ceramics, and polymers to practice using theoretical concepts to solve real-life problems.

수업방법

Lecture

학습윤리

You are encouraged to discuss course materials and concepts with your peers when you study, but the homework that you submit should be done with your own effort. Directly copying contents from other sources without proper citation is not allowed.

교과정책(방침)

Absences without penalty are permitted for specific reasons, including but not limited to: documented medical emergencies, family emergencies, and reserve forces training. If an emergency causes you to miss a graded activity (such as a midterm or final exam), you must contact the instructor immediately before the graded activity.

준비물 및 기타

N/A